Every command to perform various tasks in Big data

Map Reduce:

FTP upload jar

hdfs fs -mkdir input

hdfs fs -put NYSE.csv input/

hdfs jar myjar.jar package_name/class_name input/NYSE.csv output/

hadoop dfs -getmerge /user/cdac/output/ combined.txt

Hive

set hive.cli.print.current.db = true;

Create internal table

```
CREATE TABLE customers (
   cust_id INT,
   fname STRING,
)
ROW FORMAT DELIMITED
FIELDS TERMINATED BY ','
STORED AS TEXTFILE;
```

Load Data

LOAD DATA Local INPATH '`/home/cdacoctnov012/custs.txt`' INTO TABLE customers;

Create internal table using pre-existing table

```
CREATE TABLE nyse_copy AS
SELECT * FROM nyse;
```

Create external table

```
CREATE EXTERNAL TABLE ext_customers (
   cust_id INT, fname STRING, lname STRING, age INT, profession STRING
)
ROW FORMAT DELIMITED
FIELDS TERMINATED BY ','
STORED AS TEXTFILE
LOCATION '/user/cdacoctnov012/db';
```

Create Bucket

```
CREATE TABLE nyse_bucketed (
   stk_date STRING,
   symbol STRING
)
CLUSTERED BY (symbol) INTO 4 BUCKETS
ROW FORMAT DELIMITED
FIELDS TERMINATED BY '\t';
```

Loading Data

```
SET hive.enforce.bucketing=true;

INSERT OVERWRITE TABLE nyse_bucketed SELECT * FROM nyse;
```

Create Partition

```
CREATE TABLE nyse_partitioned (
   stk_date STRING,
   symbol STRING,
)
PARTITIONED BY (year STRING)
ROW FORMAT DELIMITED
FIELDS TERMINATED BY '\t';
```

Loading Data

```
SET hive.exec.dynamic.partition=true;
SET hive.exec.dynamic.partition.mode=nonstrict;
INSERT OVERWRITE TABLE nyse_partitioned
PARTITION (year)
SELECT
   stk_date,
   symbol,
   year(stk_date) AS year
FROM nyse;
```

ORC Table

```
CREATE TABLE txns_orc
STORED AS ORC
AS SELECT * FROM transactions;
```

Parquet Table

```
CREATE TABLE txns_parquet
STORED AS PARQUET
AS SELECT * FROM transactions;
```

```
hdfs dfs -du -h /user/hive/warehouse/nishad_one.db/txns_orc/
```

hdfs dfs -du -h /user/hive/warehouse/nishad_one.db/txns_parquet/

hdfs dfs -du -h /user/hive/warehouse/nishad_one.db/transactions

Rdds

Enter data into RDDs
rdd1=sc.textFile("/user/cdacocnov50017/input/txns2.txt")

Count
rdd1.count()

Only get selected row
rdd2=rdd1.map(lambda a: (a.split(",")[6],float(a.split(",")[3])))

To group
rdd3=rdd2.reduceByKey(lambda a,b:a+b)

To sort
rdd4=rdd3.sortBy(lambda a: -a[1])

rdd2=rdd1.map(lambda a:((datetime.strptime(a.split(",")[1],format_date).strftime("%B"),datetime.strptime(a.split(",")[1],format_date).year),float(a.split(",")[3])))

rdd4=rdd3.sortBy(lambda a: (a[0][1],datetime.strptime(a[0][0],'%B').month))

```
create the topic
--------------
$KAFKA_HOME/bin/kafka-topics.sh --create --topic training10
--bootstrap-server master:9092

list a topic
---------------
$KAFKA_HOME/bin/kafka-topics.sh --list --bootstrap-server master:9092



launch a producer console
--------------------------
$KAFKA_HOME/bin/kafka-console-producer.sh --topic training10
--bootstrap-server master:9092

launch a consume console
------------------------
$KAFKA_HOME/bin/kafka-console-consumer.sh --topic training10
--from-beginning --bootstrap-server master:9092
```